

Functional Annotation of *hutU* (Q88CZ6, PP_5033): Urocanase of *Pseudomonas putida* KT2440

Summary

The gene **hutU** (UniProt **Q88CZ6**; ordered locus **PP_5033**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes **urocanase** (urocanate hydratase; systematic name 4-imidazolone-5-propanoate hydro-lyase; EC 4.2.1.49). This enzyme catalyzes the **second step of the histidine utilization (hut) pathway**, the hydration of *trans*-urocanate to 4(5)-imidazolone-5(4)-propionate. The gene symbol, protein family (urocanase family), and domain architecture (Urocanase/Urocanase-like/Urocanase_C domains, IPR023637/IPR035401 etc.) are fully consistent with the UniProt annotation, and the identification is supported by direct experimental characterization of the *P. putida* urocanase — including cloning of the *hutU* gene, high-resolution crystallography, and detailed enzymological mechanism studies. **This is a well-characterized enzyme, and the gene identity is unambiguous.**

Mechanistically, urocanase is remarkable among lyases in that it uses a **tightly (but non-covalently) bound NAD⁺** as an **electrophilic, non-redox prosthetic group** rather than as a hydride-accepting redox cofactor. The nicotinamide ring of NAD⁺ forms a transient covalent C–C bond to the C5' position of the substrate's imidazole ring, activating the conjugated double-bond system for the stereospecific addition of water. This unusual chemistry was established through ¹³C-NMR of specifically labelled NAD⁺, deuterium isotope-effect studies, and mutational analysis identifying a single essential catalytic cysteine. The enzyme is a **cytoplasmic, soluble homodimer** of 2 × 557 residues; its 1.14 Å crystal structure (from *P. putida*) reveals an NAD-binding Rossmann domain inserted into a larger core domain that forms the dimer interface and sequesters the substrate and cofactor in a closed cavity.

Physiologically, urocanase enables *P. putida* KT2440 to use **L-histidine as a carbon, nitrogen, and energy source**. Histidine is imported by the transporter **HutT** and degraded intracellularly: histidase (HutH) converts histidine to urocanate, urocanase (HutU) hydrates urocanate to imidazolone-propionate, and downstream enzymes (HutI, then HutG/HutF) channel the product toward **L-glutamate**. The *hutU* gene lies in the urocanate-inducible

hutUHIG operon, controlled by the local repressor **HutC** (with urocanate as the physiological inducer) and integrated into global carbon/nitrogen regulation via the **CbrAB/ σ^{54}** and **NtrBC/ σ^{70}** systems. Notably, urocanate — the substrate consumed by urocanase — doubles as a **signaling molecule** through which bacteria recognize eukaryotic hosts, linking this central metabolic enzyme to environmental sensing and, in pathogens, virulence.

Gene / Protein Identity Verification

Field	Value
UniProt accession	Q88CZ6
Gene symbol	<i>hutU</i>
Ordered locus	PP_5033
Protein	Urocanate hydratase (urocanase)
EC number	4.2.1.49
Alt. name	Imidazolonepropionate hydrolase
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Family	Urocanase family
Length / MW	557 residues / $\approx$ 60.7 kDa
Cofactor	1 NAD ⁺ per subunit (tightly, non-covalently bound)
Localization	Cytoplasm
Pathway	L-histidine degradation to L-glutamate, step 2/3

Verification result: All literature located for *hutU* refers to bacterial urocanase in the histidine utilization pathway; no conflicting gene of the same symbol was found. The protein family and domain assignments align across all sources, and the *P. putida* enzyme has been directly characterized. The gene identity is secure.

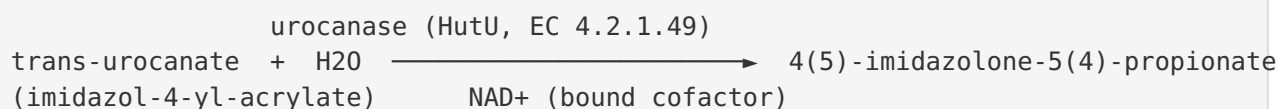
Key Findings

Finding 1 — *hutU* encodes urocanase, the second enzyme of histidine degradation

The *hutU* gene of *Pseudomonas putida* was directly demonstrated to encode a catalytically active urocanase. Lenz & Rétey (1993) isolated the *hutU* gene from a genomic library of *P. putida* nicII, subcloned it into a T7 expression vector, and expressed it in *E. coli*, where the recombinant enzyme accumulated to roughly **30% of soluble cellular protein** and was catalytically active (PMID: 7901006). The reaction catalyzed — hydration of urocanate to 4(5)-imidazolone-5(4)-propionate — and the enzyme identity (EC 4.2.1.49, NAD-dependent urocanate hydro-lyase) had earlier been established for the *P. putida* enzyme by Egan, Matherly & Phillips (1981) (PMID: 6110440).

The target protein Q88CZ6 is the KT2440 ortholog of this characterized enzyme. It belongs to the same species (*P. putida*), carries the same gene symbol, and occupies the analogous position in the *hut* pathway. Its curated reaction — *trans*-urocanate + H₂O → 4-imidazolone-5-propanoate (Rhea:13101; EC 4.2.1.49) — matches the experimentally verified chemistry exactly.

Reaction catalyzed:



Finding 2 — Urocanase uses a tightly bound NAD⁺ as a non-redox electrophilic prosthetic group

Urocanase is mechanistically distinctive: although it requires **NAD⁺**, it does not use it for a net oxidation–reduction. Egan, Matherly & Phillips (1981) used deuterium isotope effects and labeling patterns to show that the reaction proceeds by **water addition across the conjugated double-bond system of urocanate**, "rather than an internal oxidation–reduction process, yet NAD⁺ is required" (PMID: 6110440).

The role of NAD⁺ was clarified by ¹³C-NMR work of Klepp, Rétey and colleagues (1990), who grew the enzyme on [4-¹³C]nicotinate to specifically label the prosthetic NAD⁺. This revealed a **covalent C–C bond between C4 of the nicotinamide ring and C5' of the imidazole ring** of the substrate/inhibitor adduct — direct evidence for the electrophilic addition of enzyme-bound NAD⁺ to the substrate that activates it toward hydration (PMID: 1976515). A

complementary inhibition study with 2-methylurocanate proposed the same crucial first step: electrophilic addition of enzyme-bound NAD to the 2-position of the imidazole nucleus (PMID: 2885981).

Mutational analysis pinpointed the catalytic machinery: of the enzyme's cysteines, **only Cys410 was essential for catalysis**, and NAD is **not covalently bound** to the protein (PMID: 7901006). Together these results define urocanase as an enzyme in which NAD⁺ functions as a tightly bound electrophilic catalyst — a rare, non-canonical role for this classic redox cofactor.

Finding 3 — Urocanase is a cytoplasmic homodimer with an NAD-binding domain inserted into a novel core domain

The three-dimensional architecture was solved to atomic resolution. Kessler, Rétey & Schulz (2004) determined the **1.14 Å crystal structure** of *P. putida* urocanase, revealing a **symmetric homodimer of 2 × 557 amino acid residues with tightly bound NAD⁺ cofactors** (PMID: 15313616). Each subunit comprises a **typical NAD-binding (Rossmann-type) domain inserted into a larger core domain** that forms the dimer interface and binds urocanate in a surface depression.

Critically for the mechanism, the structure showed that **substrate, nicotinamide, and five water molecules are completely sequestered in a closed cavity** (PMID: 15313616). This supports the water-addition mechanism (the five ordered waters are positioned for the hydration chemistry) and implies that the NAD-binding domain acts as a mobile lid that must lift to admit substrate and release product. The soluble, cytoplasmic nature of the enzyme is consistent with its role in intracellular amino-acid catabolism.

Finding 4 — *hutU* is part of the urocanate-induced *hutUHIG* operon under global carbon/nitrogen control

In *P. putida*, the *hut* genes are contiguous, with the gene order determined as **hutG–hutI–hutH–hutU–hutC–hutF**, organized into three transcriptional units; *hutU* (with *hutHIG*) forms an operon that possesses its own operator–promoter and is **induced by urocanate** via the HutC repressor (Hu & Phillips, 1988) (PMID: 2842309). That same study showed that expression of *hutF* and *hutU* (urocanase) is induced by urocanate.

Regulation is layered onto global carbon/nitrogen homeostasis. Work in the closely related *P. fluorescens* SBW25 (Zhang & Rainey, 2007/2008) established that the *hutU* operon is transcribed from a **σ⁵⁴ promoter requiring CbrB** when histidine is the sole carbon source, and from a **σ⁷⁰ promoter requiring NtrC (or CbrB)** when histidine is the sole nitrogen source (PMID:

[18202367](#)). The three *hut* operons are **negatively regulated by the HutC repressor with urocanate as the physiological inducer** (PMID: [17717196](#)). This dual regulatory logic ensures urocanase is produced only when its substrate is available and when histidine catabolism is nutritionally advantageous.

Finding 5 — Q88CZ6 (KT2440) sequence directly matches the characterized *P. putida* urocanase, with conserved catalytic residues

Sequence-level analysis confirms that the target protein is the enzyme described above. The UniProt Q88CZ6 sequence is **exactly 557 residues** (MW $\approx$ 60.7 kDa), identical in length to the crystallized *P. putida* urocanase subunit (2×557 residues) (PMID: [15313616](#)). It contains 7 cysteines (positions 64, 192, 198, 355, 411, 508, 544); the **essential catalytic cysteine** identified in the biochemically characterized nicII enzyme (Cys410 in that numbering) (PMID: [7901006](#)) is **conserved at position 411** of Q88CZ6.

UniProt's curated annotations for Q88CZ6 are internally consistent with all experimental evidence: reaction = 4-imidazolone-5-propanoate $\rightleftharpoons$ *trans*-urocanate + H₂O (Rhea:13101, EC 4.2.1.49); cofactor = "Binds 1 NAD⁺ per subunit"; subcellular location = **Cytoplasm**; and pathway = "L-histidine degradation into L-glutamate ... step 2/3." The high similarity of urocanase across bacterial species (including psychrotrophic and mesophilic *Pseudomonas*) further supports transfer of the detailed *P. putida* characterization to the KT2440 ortholog (PMID: [11772602](#)).

Finding 6 — Histidine is imported by HutT and degraded intracellularly; urocanate is both substrate and host-recognition signal

In *P. putida* KT2440, the **APC-family high-affinity histidine:H⁺ symporter HutT** (encoded within the *hut* operon) is the major histidine uptake system; deletion of *hutT* **severely impairs growth on histidine**, indicating that histidine is imported before being catabolized intracellularly (PMID: [34245008](#)). Once inside, histidase produces urocanate, which urocanase then hydrates — all in the cytoplasm.

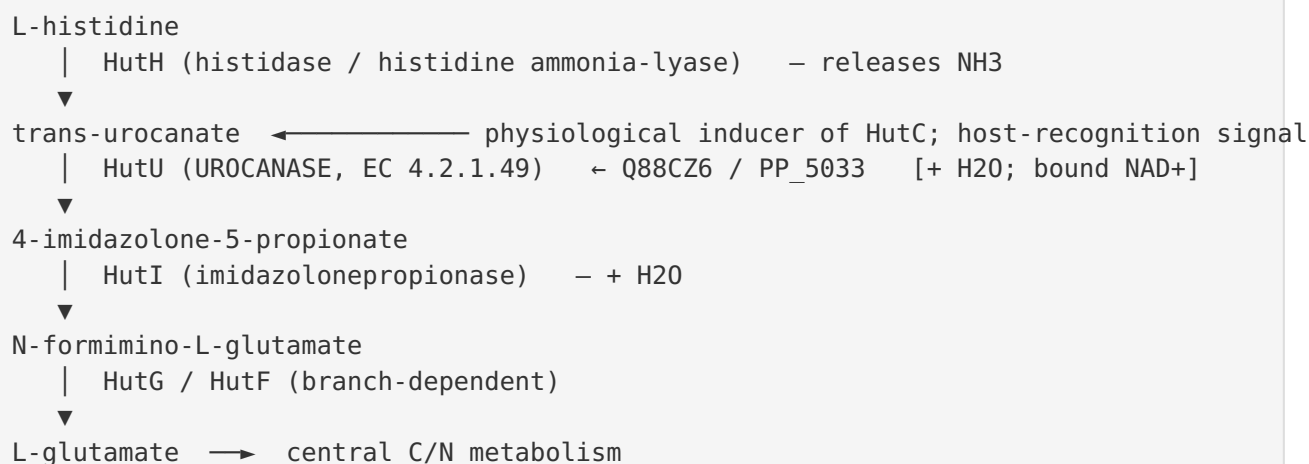
Beyond its metabolic role, **urocanate is a signaling molecule**. Zhang, Ritchie & Rainey (2014) showed that urocanate — an intermediate of the histidine degradation pathway — **accumulates in host tissues such as skin and acts as a molecule that promotes bacterial infection through interaction with the bacterial regulatory protein HutC** (PMID: [24305948](#)). Because urocanase directly consumes urocanate, it sits at the junction between

metabolism and this host-recognition/virulence signaling axis: the enzyme's activity modulates the intracellular pool of the very metabolite that serves as the HutC inducer and, in pathogens, a host-derived infection cue.

Mechanistic Model / Interpretation

Position in the histidine utilization pathway

Urocanase catalyzes step 2 of a conserved multi-step route that converts L-histidine to L-glutamate, feeding carbon and nitrogen into central metabolism:



All of these reactions occur in the **cytoplasm**, after histidine is transported into the cell by **HutT**. UniProt places urocanase at "step 2/3" of "L-histidine degradation into L-glutamate," matching this scheme.

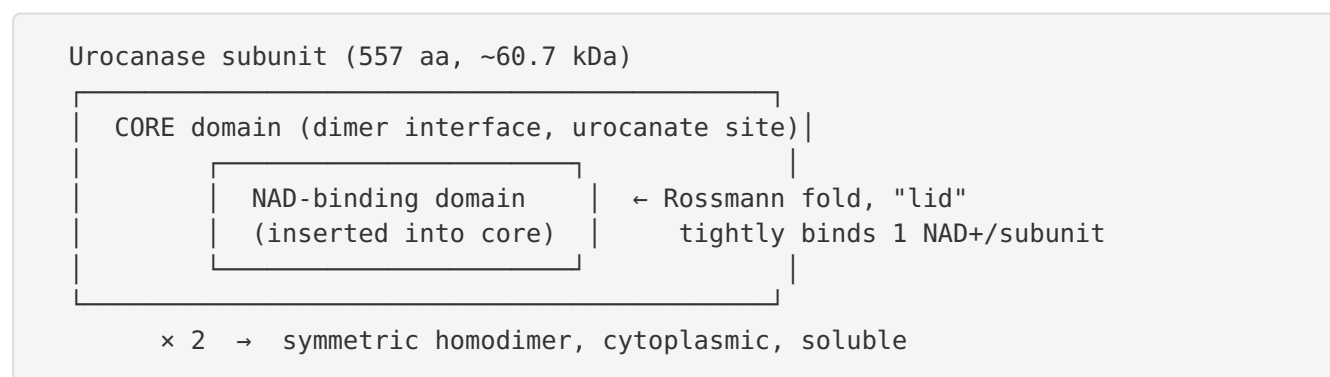
Catalytic mechanism

The core innovation of urocanase is the use of NAD⁺ as an **electrophile, not an oxidant**:

Feature	Evidence	Reference
NAD ⁺ required, but no net redox	Deuterium isotope effects; water addition, not oxidation–reduction	PMID: 6110440
Covalent C4(nicotinamide)–C5' (imidazole) adduct	¹³ C-NMR of [4- ¹³ C]nicotinate-labelled NAD ⁺	PMID: 1976515
Electrophilic addition to imidazole is first step	Inhibition by 2-methylurocanate	PMID: 2885981
Single essential Cys; NAD non-covalently bound	Site-directed mutagenesis (Cys410 essential)	PMID: 7901006
Substrate + nicotinamide + 5 waters sequestered	1.14 Å crystal structure	PMID: 15313616

In outline: the nicotinamide ring adds electrophilically to the imidazole of urocanate, forming a covalent adduct that dearomatizes/activates the conjugated acrylate-imidazole system. This allows stereospecific addition of water (delivered by the ordered active-site waters) across the double bond, after which the adduct collapses to release 4-imidazolone-5-propionate and regenerate NAD⁺. Cys410/411 participates in catalysis, and the whole cycle takes place in a closed cavity formed as the Rossmann NAD-domain lid seals over the core domain.

Structural summary



Regulatory and ecological integration

Urocanase expression is gated by substrate availability and nutritional context. The *hutUHIG* operon is de-repressed by HutC when **urocanate** accumulates, and its promoters are activated through **CbrAB**/ σ^{54} (carbon-source mode) or **NtrBC**/ σ^{70} (nitrogen-source mode). This makes

urocanase production responsive to whether histidine is being used primarily as a carbon or nitrogen source. The dual identity of urocanate — catabolic intermediate *and* host-recognition/virulence signal via HutC — means that flux through urocanase influences not just energy metabolism but also the intracellular signaling pool that couples histidine catabolism to environmental/host sensing.

Evidence Base

PMID	Study (paraphrased)	How it supports the annotation
7901006	<i>Cloning, expression and mutational analysis of hutU from P. putida</i>	Directly proves <i>hutU</i> encodes active urocanase; identifies essential Cys410; NAD non-covalent
6110440	<i>Mechanism of urocanase by deuterium isotope effects and labeling</i>	Establishes EC 4.2.1.49 identity and water-addition (non-redox) mechanism in <i>P. putida</i>
1976515	<i>Specific ¹³C-labelling of prosthetic NAD⁺; revised adduct structure</i>	Demonstrates covalent NAD–substrate C–C adduct central to electrophilic mechanism
2885981	<i>Inhibition by 2-methylurocanate; proposed mechanism</i>	Supports electrophilic addition of NAD to imidazole as the first catalytic step
15313616	<i>Structure and action of urocanase (1.14 Å)</i>	Defines homodimer (2×557), domain organization, sequestered active site with 5 waters
2842309	<i>Organization and regulation of hut genes in P. putida</i>	Establishes <i>hutG-hutI-hutH-hutU-hutC-hutF</i> order and urocanate induction of <i>hutU</i>
17717196	<i>Genetic analysis of hut in P. fluorescens SBW25</i>	HutC/urocanate negative regulation of the <i>hutU</i> -containing operon
18202367	<i>Dual CbrAB/NtrBC regulation of hut</i>	σ^{54} /CbrB (carbon) vs σ^{70} /NtrC (nitrogen) control of the <i>hutU</i> operon
34245008	<i>HutT is the major L-histidine transporter in KT2440</i>	Establishes histidine import route feeding intracellular urocanase
24305948	<i>Urocanate as a host-recognition signal</i>	Shows urocanate (urocanase substrate) is a HutC-acting signaling molecule
11772602	<i>Cold-inducible hutU from P. syringae</i>	Confirms high urocanase sequence conservation and Rossmann/NAD motif across <i>Pseudomonas</i>

Supporting/contextual literature includes the demonstration that the *hut* operon is upregulated at low temperature in antarctic *Pseudomonas* (PMID: [9561727](#)), the branched *hut* pathway topology and horizontal acquisition of *hutE* in *P. aeruginosa* PAO1 (PMID: [22225844](#)), CbrA's

dual role in histidine uptake and signaling (PMID: 26148710), HutC governance of NtrBC carbon/nitrogen homeostasis (PMID: 33675669), and the identification of urocanase homology within the *Bacillus subtilis hut* operon (PMID: 7704263).

Consistency check: No literature was found describing a *different* gene with the symbol *hutU*. In every reference, *hutU* refers to bacterial urocanase in the histidine utilization pathway, and the protein family/domain assignments (urocanase family; Rossmann NAD-binding fold; conserved active-site cysteine) align across all sources. The gene identity is therefore secure.

Limitations and Knowledge Gaps

- 1. Direct characterization is of *P. putida* strains other than KT2440.** The definitive enzymology, mutagenesis, and 1.14 Å structure were performed on *P. putida* (nicII / the strain used for crystallography), not on KT2440 itself. The inference to Q88CZ6 rests on identical species, identical protein length (557 aa), conserved catalytic cysteine (position 411), and shared gene symbol/operon context — very strong, but formally by orthology rather than direct KT2440 assay.
 - 2. Catalytic-residue numbering.** The "Cys410" essential cysteine of the nicII enzyme corresponds to position 411 in Q88CZ6; a full residue-by-residue alignment confirming all active-site residues (beyond the catalytic Cys) was not exhaustively documented here.
 - 3. Kinetic parameters for the KT2440 enzyme** (k_{cat}, K_m for urocanate, substrate-specificity profile against related imidazole-acrylates) are not reported in the reviewed literature for Q88CZ6 specifically.
 - 4. Substrate specificity breadth.** Urocanase is understood to be highly specific for *trans*-urocanate, but quantitative specificity data (e.g., activity toward analogues beyond the 2-methylurocanate inhibitor) were not systematically compiled.
 - 5. In-vivo flux and physiological knockout data in KT2440.** While *hutT* deletion phenotypes are documented, a clean *hutU*-specific deletion phenotype in KT2440 (growth on histidine, urocanate accumulation) was not directly cited.
 - 6. Localization** is annotated as cytoplasmic (UniProt) and inferred from the soluble homodimeric structure; no experimental fractionation study specific to KT2440 was reviewed.
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Proposed Follow-up Experiments / Actions

1. **Recombinant KT2440 urocanase characterization.** Express Q88CZ6 (PP_5033) in *E. coli*, purify, and measure steady-state kinetics (kcat, Km for *trans*-urocanate), NAD⁺ stoichiometry, and thermal stability to confirm transfer of the nicII/crystallography data to the exact target sequence.
 2. **Active-site alignment and mutagenesis.** Perform a structure-guided alignment of Q88CZ6 against the 1.14 Å structure (PMID: 15313616); verify conservation of all substrate- and NAD-contacting residues, and validate Cys411 as the catalytic cysteine by C411A/S mutagenesis.
 3. **KT2440 hutU deletion phenotype.** Construct a markerless Δ hutU mutant; assay growth on histidine as sole C/N source, and measure intracellular urocanate accumulation (expected to rise) and its effect on HutC-dependent gene expression.
 4. **Metabolomic confirmation of product.** Use LC-MS/NMR to confirm 4-imidazolone-5-propionate as the *in vivo* product and to trace flux toward L-glutamate through HutI/HutG/HutF.
 5. **Regulatory dissection in KT2440.** Test σ^{54} /CbrB vs σ^{70} /NtrC promoter usage of the *hutU* operon directly in KT2440 under carbon- vs nitrogen-limited histidine growth, mirroring the SBW25 studies.
 6. **Signaling link.** Quantify how urocanase activity modulates the intracellular urocanate pool and downstream HutC-controlled/virulence-associated gene expression, to test the metabolism–signaling coupling suggested by PMID: 24305948.
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Conclusion

hutU (Q88CZ6, PP_5033) of *Pseudomonas putida* KT2440 encodes **urocanase (EC 4.2.1.49)**, the second enzyme of the histidine utilization pathway, which hydrates *trans*-urocanate to 4-imidazolone-5-propionate using a tightly bound, non-covalent NAD⁺ acting as an electrophilic (non-redox) catalyst. It is a soluble, cytoplasmic homodimer (2 × 557 residues) with a fully solved atomic-resolution structure, and it functions to let the bacterium exploit histidine (imported by HutT) as a carbon, nitrogen, and energy source by routing it to L-glutamate. Its

expression is urocanate-induced through the HutC repressor and integrated into global C/N regulation via CbrAB/ σ^{54} and NtrBC/ σ^{70} . The gene identity is unambiguous and strongly supported by direct experimental evidence in *P. putida*.

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